$\Sigma_{\mathrm{E}}$ or, more generally, a distribution. For a canonical system there is no longer a constraint of the form $H = \mathrm{E}$ and thus we must allow the phase space to include essentially all points for which the position coordinates lie in the defined configurational domain (a 3-torus, if periodic boundary conditions are used). In general the momenta are unbounded. It thus becomes necessary to restrict the set of functions on which the functional $\langle \cdot, \rho \rangle$ acts. In the case of periodic boundary conditions we will assume that, for any observable of interest,

$$|\phi(\boldsymbol{q},\boldsymbol{p})| = \mathcal{O}(\|\boldsymbol{p}\|^{\alpha}), \qquad \boldsymbol{p} \to \infty,$$

for some positive constant $\alpha$.

If the position domain $\mathcal{Q}$ is unbounded (for example choosing $\mathcal{Q} = \mathbb{R}^{\mathrm{N_c}}$) then some other restriction is needed on the potential energy function $U$ (and on the observables) in order that averages are well defined. Typical assumptions for potentials on the unbounded domain are that (i) $U$ is bounded below, and (ii) $U$ grows sufficiently rapidly as $\boldsymbol{q} \to \infty$. This ensures that

$$\int_{\mathcal{Q}} \exp(-\beta U(\boldsymbol{q}))\mathrm{d}\omega_q < \infty.$$

Under assumptions on the potential energy as discussed above, and with $\phi$ continuous and allowed to grow at most polynomially fast at infinity, we may proceed to define averages with respect to the canonical density $\rho_\beta$, namely

$$\mathrm{Av}_\beta(\phi) = Z_{\mathrm{can}}^{-1} \int_{\mathcal{D}} \phi(\boldsymbol{q},\boldsymbol{p})\rho_\beta(\boldsymbol{q},\boldsymbol{p})\mathrm{d}\omega. \tag{6.4}$$

### 6.1.5 Calculation of Temperature

Consider the one degree-of-freedom model $H(q,p) = \frac{p^2}{2m} + U(q)$ on the unbounded domain, with $U$ bounded below and growing sufficiently rapidly as $|q| \to \infty$. The associated canonical density is

$$\rho_\beta = Z_{\mathrm{can}}^{-1} \exp\left(-\beta\left[\frac{p^2}{2m} + U(q)\right]\right),$$

where

$$Z_{\mathrm{can}}^{-1} = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \exp\left(-\beta\left[\frac{p^2}{2m} + U(q)\right]\right)\mathrm{d}p\mathrm{d}q.$$